

3rd_Ch5_Example 5-2

Use the **time and location indices** below to calculate the **forecast cost** for a **building** with **52,000 square feet of floor area**. The building is to be constructed **3 years from now** in **City B**. A **similar type building** with **41,000 square feet** of floor area that cost **\$4,510,000** was completed **2 years ago** in **City D**.

使用下方的时间和地点指数计算一栋建筑的预计成本，该建筑的地面面积为 52,000 平方英尺，计划在 3 年后在 B 市建造。一个类似的建筑，成本为 \$4,510,000，面积为 41,000 平方英尺，于 2 年前在 D 市建成。

Time	Index	Location	Index
4 yr ago	3780	City A	1251
3 yr ago	3821	City B	1372
2 yr ago	3915	City C	1452
1 yr ago	4150	City D	1589
Current year	4287	City E	1426

There is no method to predict future cost escalation with absolute certainty. For this example a compound interest is calculated based on the trend in cost indices over the past 4 years. Using the equation compound interest equation $F/P = (1 + i)^n$, an interest rate can be calculated as follows:

$$\frac{4287}{3780} = (1 + i)^4 \rightarrow i = 3.2\%$$

The combined adjustments of costs for time, location and size can be calculated as follows:

Estimated cost	Previous cost	Time adjustment	Location adjustment	Size adjustment
"	= \$4,510,000	× (1 + 0.032) ⁵	× (1372/1589)	× (52,000/41,000)
"	= \$4,510,000	× 1.17057	× 0.86344	× 1.26829
"	= \$5,781,289			